

MICROECONOMICS

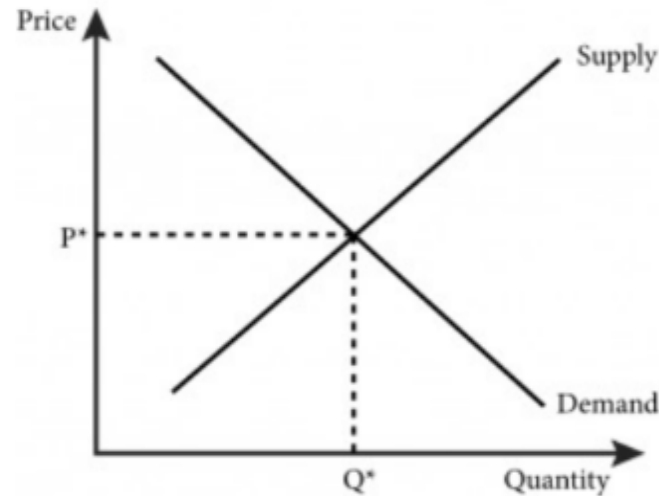
Market Forces

The market forces of Demand and Supply play an important role in the determination of equilibrium price and output in free markets.

Law of Demand : Ceteris Paribus, when the price of good falls, the quantity demanded increases and vice-versa.

Exceptions!!

Law of Supply: Other things remaining constant, when the price of good falls, the quantity supplied falls and vice-versa.



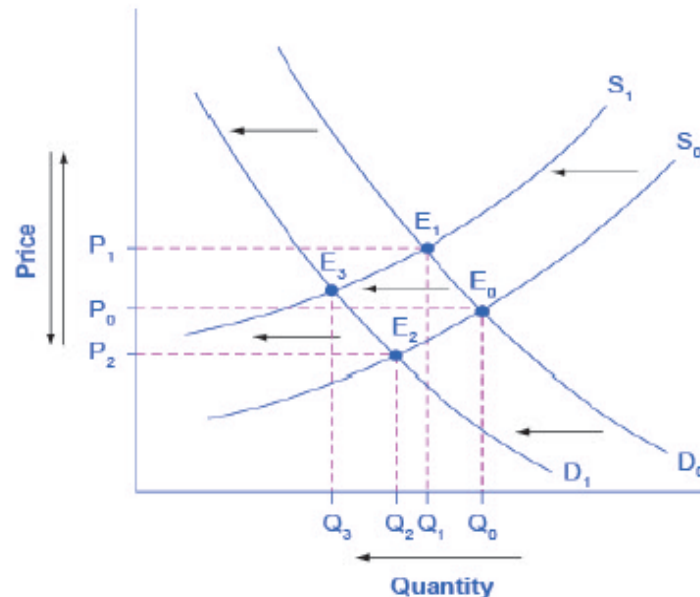
Shifts in Demand and Supply Curves

Equilibrium price and quantity is determined by the intersection of the demand and supply curves.

A fall in quantity demanded of a commodity causes the demand curve to shift inward (left) and a rise in quantity demanded causes the demand curve to shift outward (right).

An increase in quantity supplied of a commodity causes the supply curve to shift outward and vice-versa.

The new equilibrium price and output is obtained when the shifted curves intersect.



Elasticity of Demand and Supply

Elasticity of demand (E_d) is the degree of responsiveness of quantity demanded of a commodity to a change in its price. Similarly, elasticity of supply (E_s) measures the change in quantity supplied of a commodity with respect to a change in its price.

E_d is the ratio of percentage change in quantity demanded to the percentage change in its price.

$$E_d = \frac{\Delta Q}{\Delta P} * \frac{P}{Q}$$

Arc Elasticity of Demand (AE_d) uses the average change in price and the average change in quantity to calculate the elasticity.

$$AE_d = \frac{Q_2 - Q_1}{(Q_1 + Q_2)/2} * \frac{(P_1 + P_2)/2}{P_2 - P_1}$$

Elasticity of Demand and Supply

Cross Elasticity of demand (CE_d) is the degree of responsiveness of quantity demanded of a commodity to a change in the price of a related good.

If the related good is a substitute, the CE_d will be positive and if the related good is a complement, then CE_d will be negative.

$$CE_d = \frac{\% \text{ change in quantity demanded}}{\% \text{ change in the price}}$$

$$CE_d = \frac{\Delta Q}{\Delta P_{\text{related good}}} * \frac{P_{\text{related good}}}{Q}$$

Income Elasticity of Demand (IE_d) is the change in quantity demanded of a commodity with respect to a change in the consumer's income.

$$IE_d = \frac{\Delta Q}{\Delta I} * \frac{P}{I}$$

Elasticity of Demand and Supply

If, Elasticity of demand (E_d) = 0, then the quantity demanded does not respond to the change in price and the good will have a **perfectly inelastic** demand.

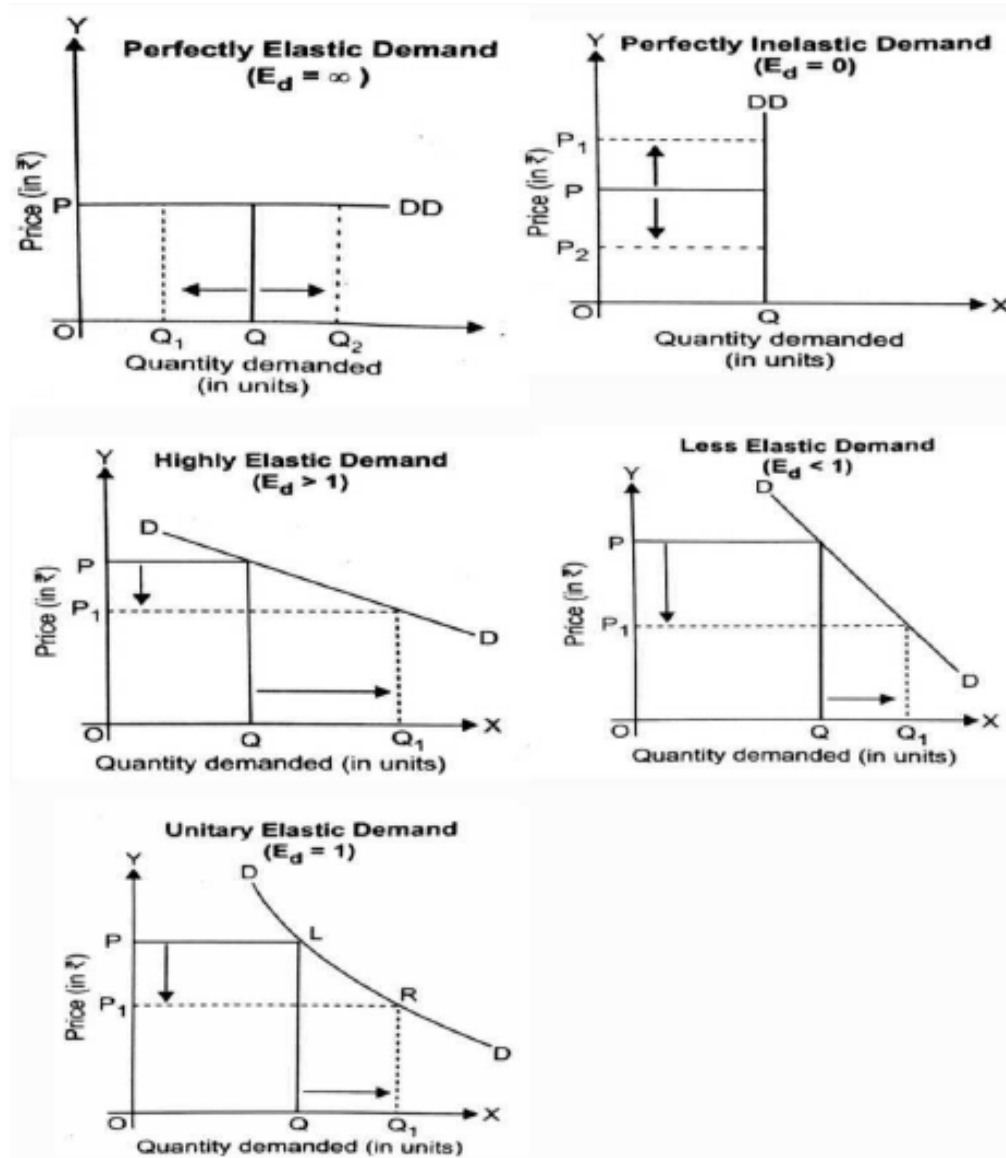
If $E_d < 1$, then the quantity demanded of a commodity responds less than proportionately to the change in its price; it will have a **relatively inelastic** demand curve.

If $E_d = 1$, then the quantity demanded of a commodity responds equally and proportionately to a change in its price; thus it will have a **unitary elastic** demand curve.

If $E_d > 1$, then the quantity demanded of a commodity responds more than proportionately to a change in its price; it will have a **relatively elastic** demand curve.

If $E_d = \infty$, then, at a given price the quantity demanded of a commodity is infinite and the demand curve is **perfectly elastic**.

Elasticity of Demand and Supply

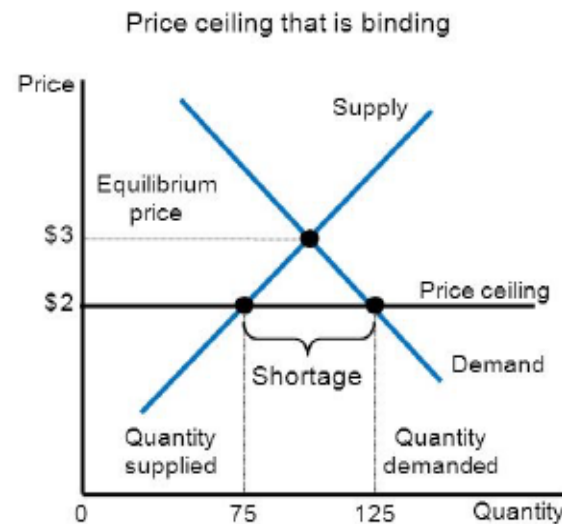
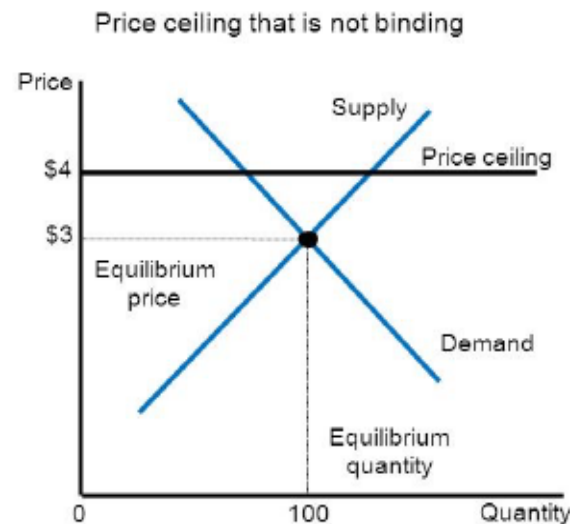


Elasticity and Price Controls

The elasticity of demand and supply plays an important role in determining the outcome of the price controls.

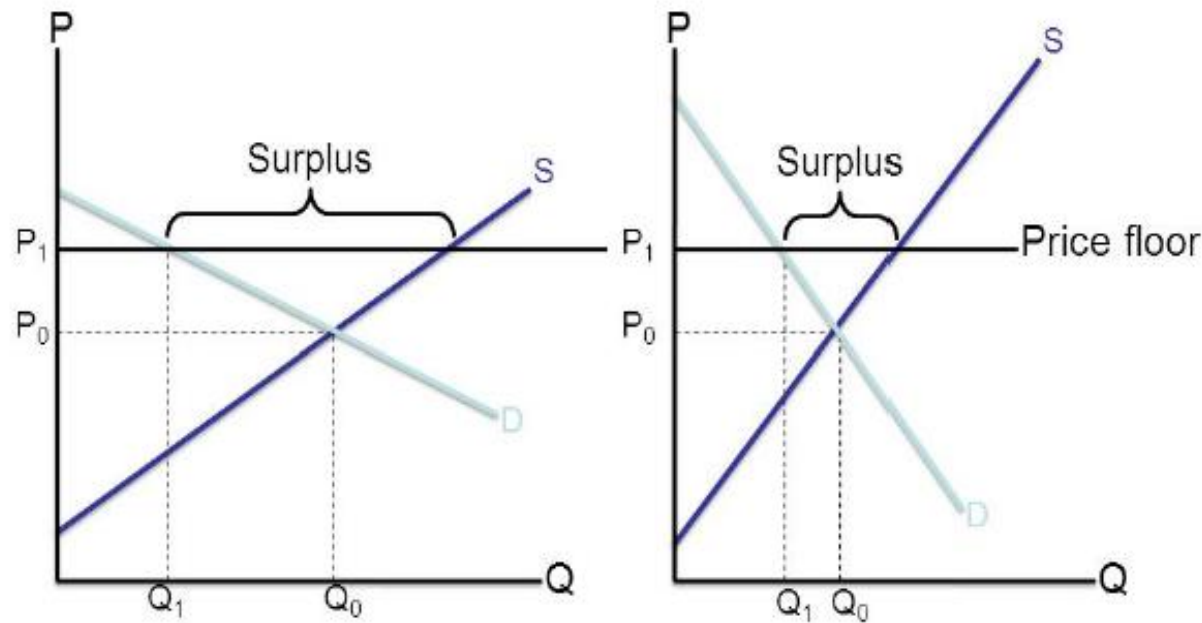
Price Ceiling is the legally established maximum price at which a good can be sold and a **Price floor** is the legally established minimum price at which a good can be sold.

If the **ceiling is set above the market equilibrium**, then it is not binding. But, if the **ceiling is set below the equilibrium**, there will be an excess demand and hence shortage of the commodity. The extent of the shortage depends upon the respective elasticities; which again depends on the nature of the commodity.



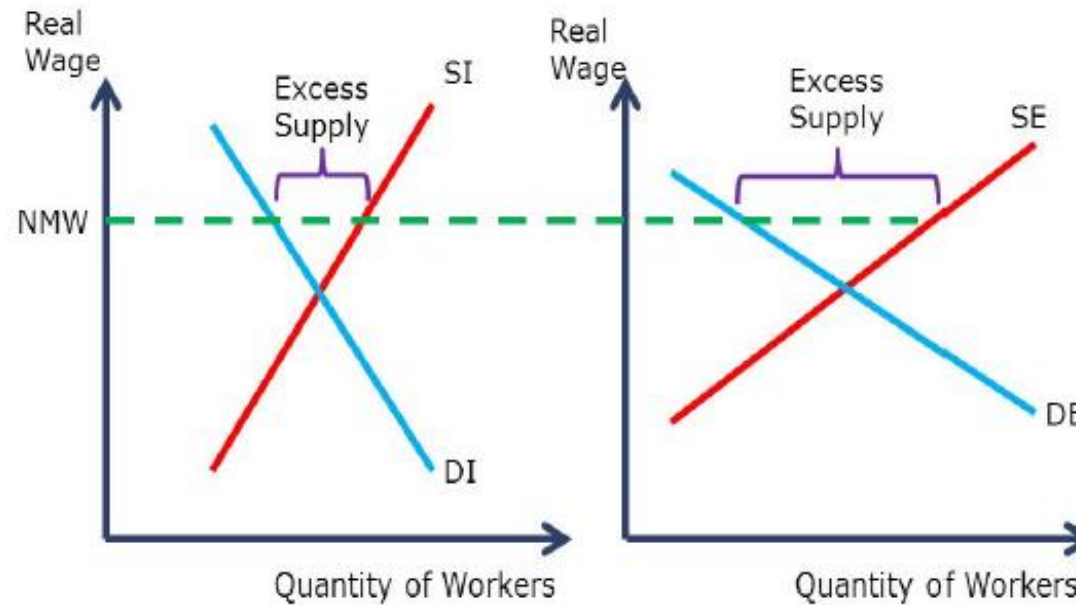
Elasticity and Price Controls

If the **price floor** is set **below the market equilibrium**, it will not be binding. But, if the **price floor** is set **above the market equilibrium**, then there will be excess supply or surplus of the commodity. Again, the extent of surplus depends upon the nature of the commodity.



Elasticity and Price Controls

If a minimum wage legally established in the labour market where the demand and supply are both relatively inelastic, then the excess supply of labour shall be lesser compared to the situation when the demand and supply are both relatively elastic.



THANK YOU